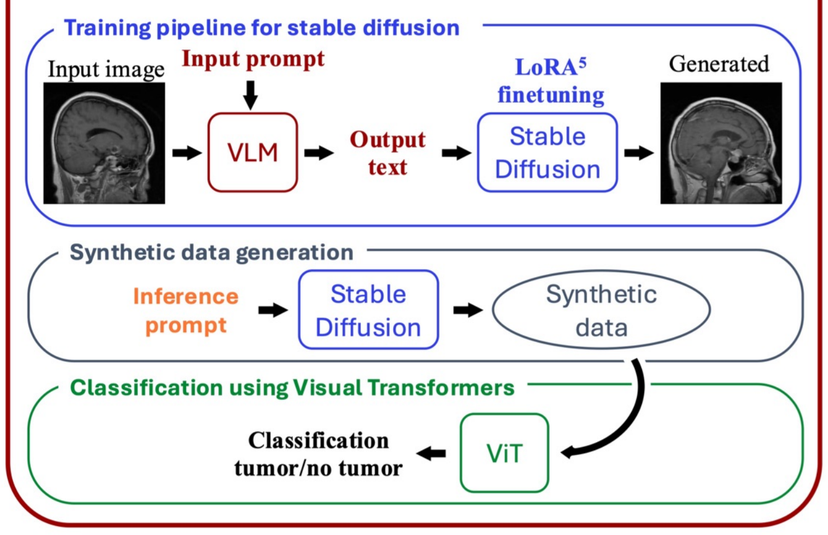
DTU Compute
Department of Applied Mathematics and Computer Science

# Generative Modeling of High Fidelity Brain Tumor MRI Images Using Vision Language & Stable Diffusion Models

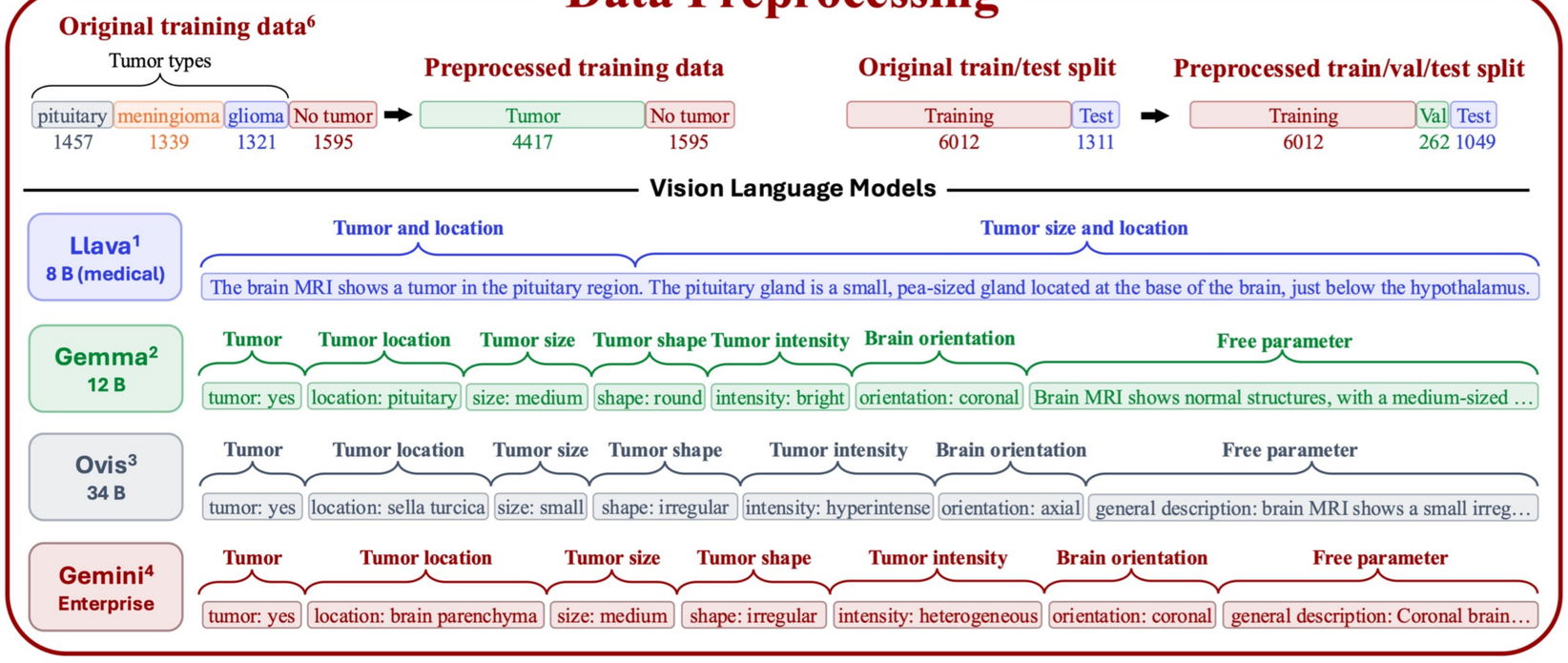
**Jone Steinhoff** , **Lukas Rasocha** , **Mads Prip** & **Petr Boska Nylander**

## Introduction

The project aims to generate synthetic MRI tumor images with controllable tumor locations via Stable Diffusion prompts, and evaluate quality using FID, KID, and a classification task.




## Data Preprocessing

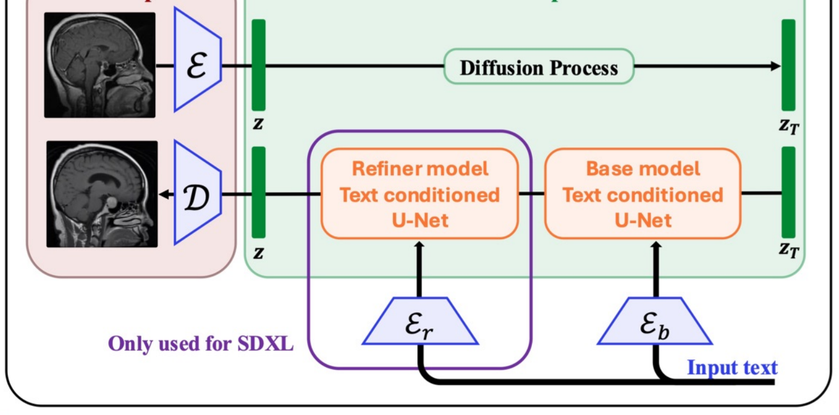



## Models

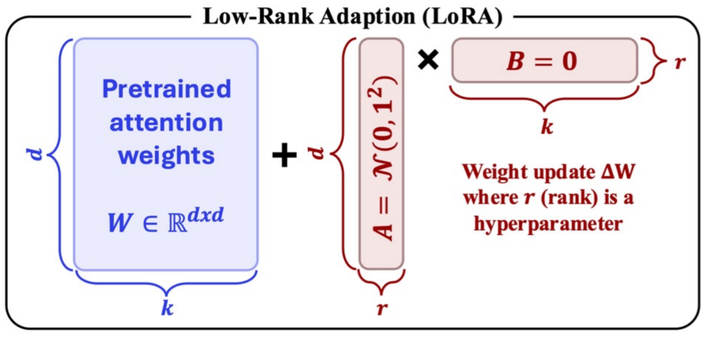
### Stable diffusion XL hyperparameters

- Rank **128/248**
- Resolution **512**
- Batch Size **2**
- Accumulation Steps **8**
- Learning Rate **1e-4**
- Gradient Checkpointing **True**
- Learning rate Scheduler **Cosine**
- SNR Gamma **5**
- Adam Weight Decay **0.01**
- Learning Rate Warmup Steps **1000**

### Stable diffusion and stable diffusion XL




### Low-Rank Adaption (LoRA)




### Vision Transformers hyperparameters

- Batch Size **16**
- Image Size **512**
- Patch Size **16**
- Channels **3**
- Embedding Dimension **256**
- Number of Heads **8**
- Number of Layers **4**
- Number of Classes **2**
- Positional Encoding **Learnable**
- Pooling Method **CLS**
- Dropout Rate **0.3**
- Fully Connected Dimension **512**
- Number of Epochs **40**
- Learning Rate **1e-4**
- Warmup Steps **625**
- Weight Decay **1e-3**
- Gradient Clipping **1.0**

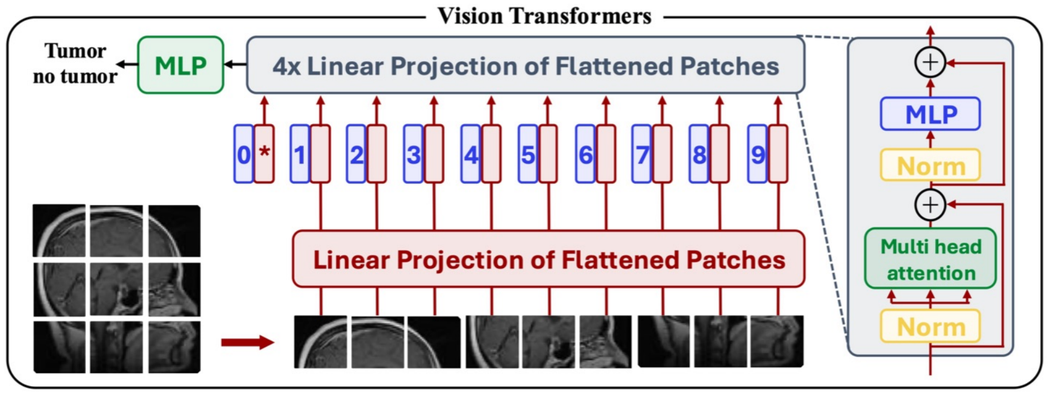
### Stable diffusion hyperparameters

- Rank **128/248**
- Resolution **512**
- Batch Size **2**
- Accumulation Steps **8**
- Learning Rate **1e-4**
- Gradient Checkpointing **True**
- Learning Rate Scheduler **Cosine**
- SNR Gamma **5**
- Adam Weight Decay **0.01**
- Learning Rate Warmup Steps **500**

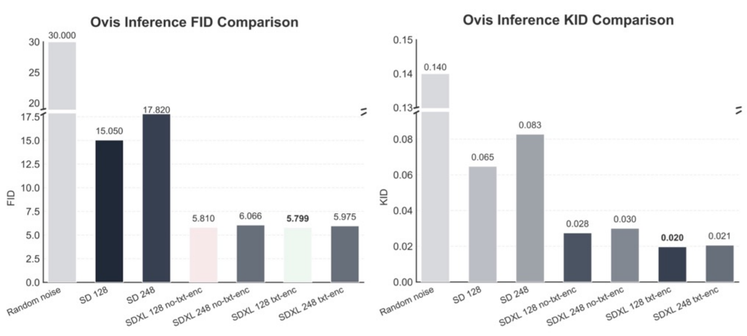
### Stable diffusion inference

- Number of Denoising Steps **50**

### Vision Transformers

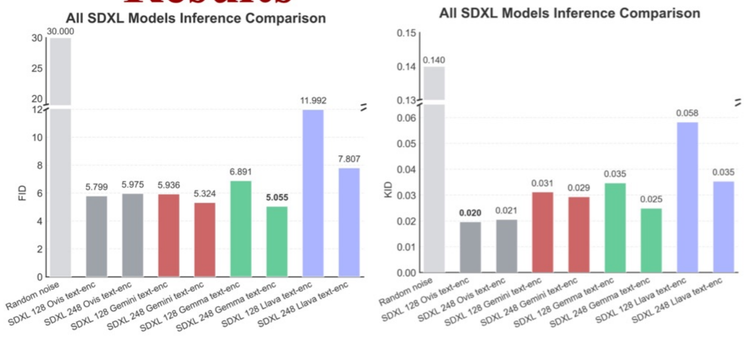



## Results




The figures presents the KID and FID scores for SD and SDXL models, both with and without a trainable text encoder, using LoRA ranks of 128 and 248.

The best-performing model is the SDXL with trainable text encoder and therefore the other VLMs are trained on this







The figures presents the KID and FID, scores for SDXL models, using LoRA ranks of 128 and 248.

The overall best-performing model is Ovis (based on KID) and Gemma (based on FID), which incorporates a trainable text encoder and a LoRA rank of 128 and 248 respectively. Synthetic data is generated using this model and employed to classify tumor versus no tumor, in comparison to the original dataset.

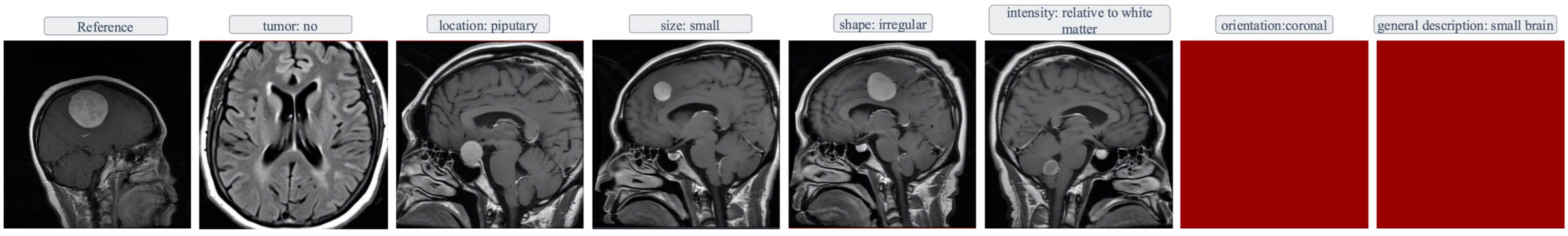


The figure show the test accuracy comparing original data and synthetic data

**Ovis Reference Prompt**

Tumor | Tumor location | Tumor size | Tumor shape | Tumor intensity | Brain orientation | Free parameter

tumor: yes | location: sella turcica | size: small | shape: irregular | intensity: hyperintense | orientation: axial | general description: Brain MRI shows a possible pituitary tumor with mixed intensity. The sella turcica appears enlarged. No other abnormalities




| Model Accuracies | No tumor | Location | Size | Shape | Intensity | Orientation |
|---|---|---|---|---|---|---|
| Ovis | 1 | 0.66 | 0.65 | 0.5 | 0.8 | 0.35 |
| Llava | 1 | 0.45 | 0.05 | 0 | 0.05 | 0 |
| Gemini | 1 | 0.82 | 0.65 | 0.7 | 0.8 | 1 |
| Gemma | | | | | | |

## Conclusions

- Combining automatic data labeling with vision-language models with fine-tuning large diffusion models shows strong potential in domains with limited data availability.
- The choice of VLM impacts the quality of generated images, where the best VLM in our set-up was **Gemma 12B** together with **SDXL**
- Using detailed and structured medical prompts to control the generation worked well, where Gemini prompts were followed most closely

## References

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2. https://huggingface.co/google/gemma-3-12b-it
3. https://huggingface.co/AIDC-AI/Ovis2-34B
4. https://ai.google.dev/gemini-api/docs/models
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